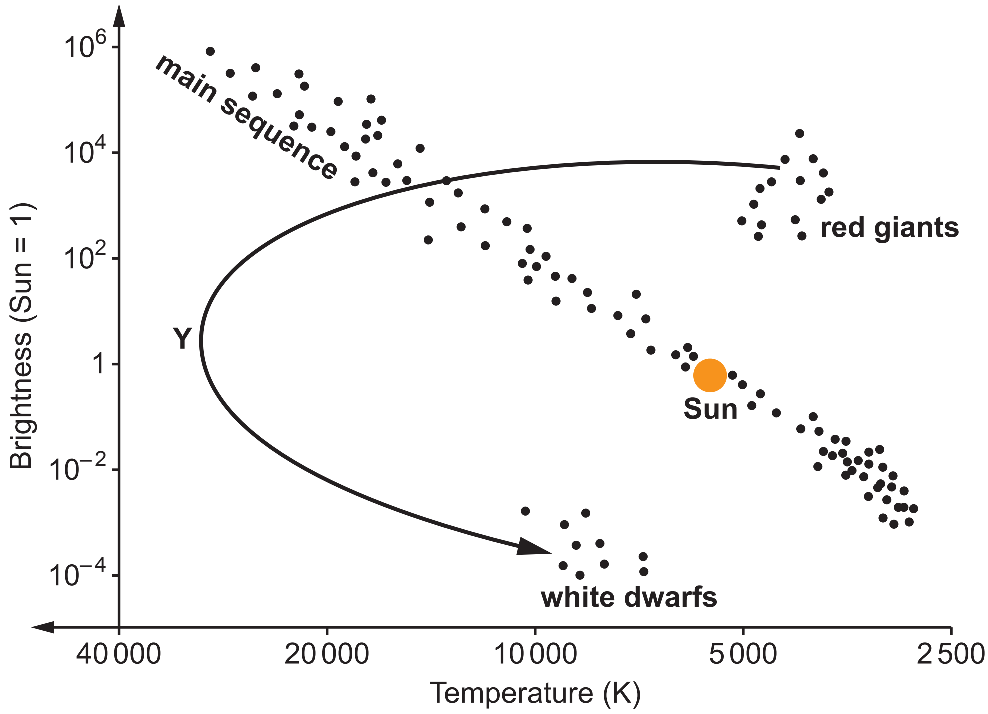
Examiner only

5. The HR diagram below shows the evolutionary path of our Sun.



(a) Currently the Sun is part of the main sequence but when it evolves into a red giant, its temperature decreases and its brightness increases.

(i) The arrow shows the path the Sun will take as it evolves from a red giant to a white dwarf.
Use information from the HR diagram, to describe how the **temperature** of the Sun changes along this path.
Use the point **Y** in your answer to help with your description. [2]

....

....

....

(ii) Use information from the HR diagram, to describe how the **brightness** of the Sun changes when it evolves from a red giant into a white dwarf. [1]

....

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(b) The life cycle of a high mass star is different from that of the Sun. Starting with the main sequence, state, in order, the remaining stages in its life cycle. [3]

....

....

....

(c) An information table obtained from the internet shows some distances used in astronomy.

Distance	Equivalent
1 light year (l-y)	63 241 astronomical units (AU)
1 astronomical unit (AU)	1.5×10^{8} km

Sirius B is a white dwarf. It is **8.6 light years** away from Earth.
Use information from the table to calculate the distance between Earth and Sirius B.
Give your answer **in metres**. [3]

distance = ... m

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